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RESEARCH ARTICLE

# Genetic structure of a disjunct peripheral population of mountain sucker *Pantosteus jordani* in the Black Hills, South Dakota, USA

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**Abstract** A peripheral population of mountain sucker, *Pantosteus jordani,* located in the Black Hills of South Dakota, USA, represents the eastern-most range of the species and is completely isolated from other populations. Over the last 50 years, mountain sucker populations have declined in the Black Hills, and now only occur in 40 % of the historic local range, with densities decreasing by more than 84 %.We used microsatellite DNA markers to estimate genetic diversity and to assess population structure across five streams where mountain suckers persist. We evaluated results in the context of recent ecological surveys to inform decisions about mountain sucker conservation. Significant allele frequency differences existed among sample streams (Global $F_{ST} = 0.041$) but there was no evidence of isolation by distance. Regionally, genetic effective size, $N_e$, was estimated to be at least 338 breeding individuals, but $N_e$ within streams was expected to be less. Despite almost complete demographic isolation and reduced population size, there appears to be little evidence of inbreeding, but genetic drift and local isolation due to fragmentation probably best explains genetic structure in this peripheral mountain sucker population. Recommended strategies for population enhancement include restoration of stream connectivity and habitat improvement. Moreover, repatriation and assisted movement (i.e., gene flow) of fishes should maximize genetic diversity in stream fragments in the Black Hills region.

**Keywords** Genetic drift · Genetic effective size · Fragmentation · Fish · Streams · Microsatellite · Catostomidae

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## Introduction

Introduction of non-native species, over-exploitation, and habitat degradation have been implicated in range reductions and declines in local abundance of many native North American freshwater fishes (Miller et al. 1989; Williams et al. 1989; Jelks et al. 2008). It is possible to conserve and restore ranges and abundances of native fishes by reversing or removing negative anthropogenic effects, but in some contexts, further remediation is needed. Managers are increasingly charged with reintroducing native fishes to sites from which the species was extirpated, initiating captive breeding programs and releasing progeny to bolster local densities, or translocation of individuals to artificially reconnect historically continuous populations. In these cases, genetic characterization of a population is a necessary step to help mitigate anthropogenic inbreeding depression or negative effects of recruitment over-fishing of persisting populations (Moran 2002; DeSalle and Amato 2004).

Mountain sucker, *Pantosteus jordani,* occurs throughout the western United States and Canada from southern

Saskatchewan and Alberta in the north (Saskatchewan River drainage), south and west to eastern California (Pacific drainages), and east through the Intermountain region to western Nebraska and the Black Hills of South Dakota (upper Missouri River drainage; Scott and Crossman 1973; Unmack et al. 2014). A peripheral population located in the Black Hills represents the eastern-most range of the species, and because streams in this region are part of the Missouri River drainage, the Black Hills population is completely isolated from other populations across the species' range. Bailey and Allum (1962) attributed mountain sucker isolation in the Black Hills to either retreat of the Pleistocene glaciers or capture of the Little Missouri River by the Belle Fourche River. In other portions of the range, mountain sucker habitat includes cool mountain streams, larger rivers, and lakes; but in the Black Hills, mountain sucker is found exclusively in mountain streams (Schultz and Bertrand 2012). The species is secure at present (G5, Natureserve 2010), but is declining at finer spatial scales, specifically in Eastern California (Moyle and Vondracek 1985; Decker 1989), Wyoming (Patton et al. 1988), and in South Dakota (Species of Greatest Conservation Need; Schultz and Bertrand 2012).

Stream sampling in the 1960s indicated that mountain sucker was abundant and widely distributed in streams in of the Black Hills (South Dakota Game, Fish and Parks unpublished data). More contemporary studies documented a range contraction of more than 60 % relative to this baseline, from approximately 600 river km in the 1960s to just under 160 river km recently (Schultz and Bertrand 2012). Across most streams occupied in the northern Black Hills, catch per unit effort (CPUE) per square meter of stream ranged from 0.01 to 0.1 fish $m^{-2}$ in the 1960s and ranged from 0.00001 to 0.0001 fish $m^{-2}$ in recent sampling events. Streams within the central portion of the Black Hills maintain denser populations (CPUE greater than 0.01 mountain suckers per $m^2$), whereas the species is less dense (CPUE less than 0.01 mountain suckers per $m^{-2}$) on the periphery of the Black Hills (Schultz and Bertrand 2012).

Mountain suckers in Black Hills streams are now isolated from one another by warmer, lower-gradient stream segments that are inhabited by introduced piscivorous brown trout that may limit movement and local abundance. Additionally, adjacent watersheds often are disconnected by dry stream reaches (termed the "loss zone" by Williamson and Carter 2001; Fig. 1). Small, isolated populations are susceptible to pronounced effects of genetic drift, which is expected to reduce within-stream genetic variation and increase genetic divergence between streams (Lesica and Allendorf 1995). Hence, each stream could harbor unique elements of the mountain sucker genome. If allele frequencies differ significantly among streams and divergent and unique alleles (i.e., private alleles) are present in streams as a result of historical isolation events (e.g., vicariance), then biologists should manage each stream as a discrete unit to conserve genetic diversity across the region (i.e., stock concept; Booke 1981; Waples and Gaggiotti 2006). Alternatively, recent disturbances (i.e., introduced species or groundwater depletion) could have resulted in fragmentation of a formerly contiguous population. Random genetic drift and isolation would lead to differentiation among subpopulations and reduce genetic diversity. In this case, subpopulations should be managed as a single population to preserve remaining levels of genetic diversity.

Mountain sucker genetic diversity and spatial genetic structure has not been previously assessed in the Black Hills, and the extent of gene flow among localities has not been evaluated. Explicit knowledge of population structure in this region will have direct implications for conserving and recovering this species in South Dakota. Therefore, our objectives were to: (i) estimate genetic diversity within and among currently occupied streams in the Black Hills region and (ii) assess population genetic structure and gene flow throughout the species range in the Black Hills of South Dakota. In addition, we followed an approach similar to Paz-Vinas et al. (2013) and considered genetic results in light of contemporary demographic information to make inferences about levels of genetic diversity and connectedness to better inform mountain sucker recovery efforts in this disjunct peripheral population.

## Materials and methods

### Fish sampling methods

Mountain suckers were sampled from currently occupied sites in the Black Hills of South Dakota, including sites in tributaries of the Belle Fourche River (Annie Creek, Bear Butte Creek and Whitewood Creek), and tributaries to the Cheyenne River (Boxelder Creek, Elk Creek, and the North Fork of Rapid Creek; Fig. 1). Fish were collected using backpack electrofishers (Smith-Root Model 12-B) with pulsed-DC current and dipnets. Pectoral fins were clipped (<5 $mm^2$ genetic tissue sample), placed in individually labeled vials, and preserved in 95 % ethanol. After biopsy, fish were recovered in fresh water and released at the site of capture.

### Molecular methods

Genomic DNA was extracted from fin tissue samples using a 96-well modification of the Wizard Genomic DNA purification kit (Promega Corp, Madison, WI). Extracted DNA concentration was quantified using a Nanodrop® ND-

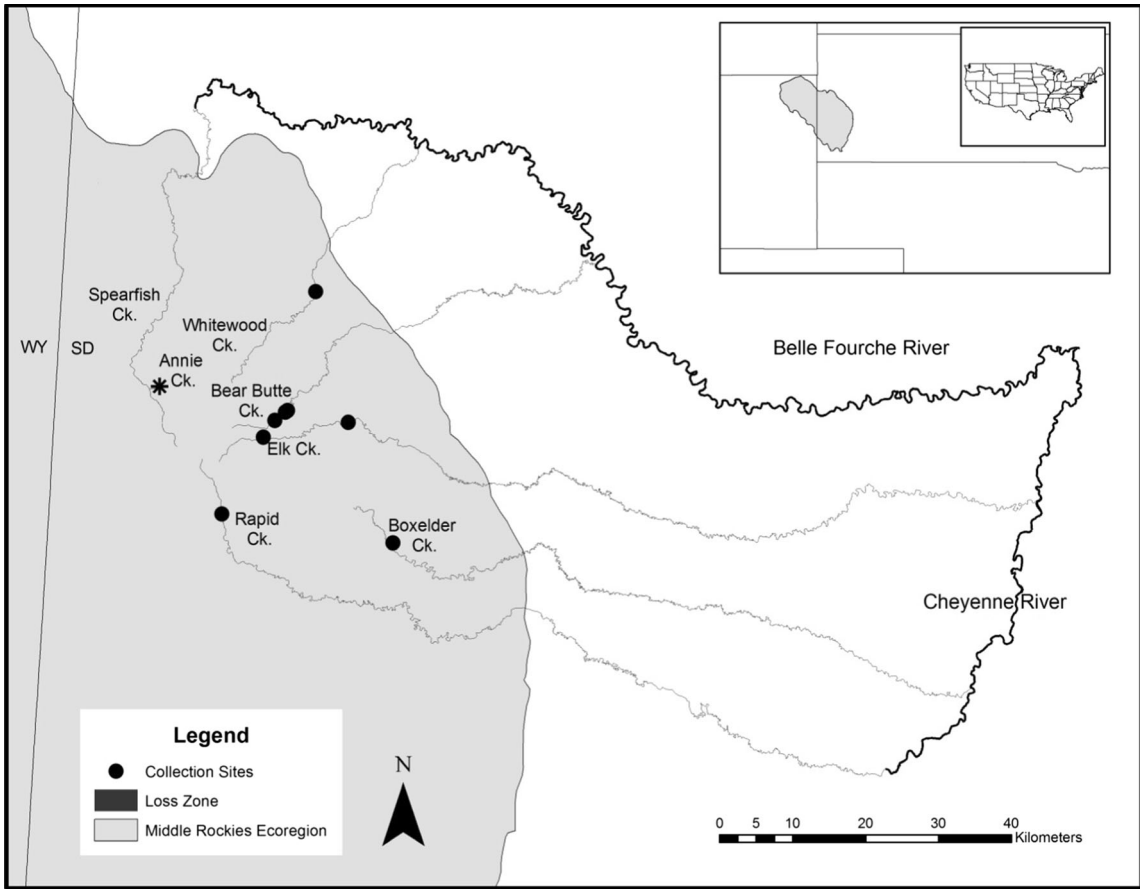


**Fig. 1** Map of study sites from which mountain sucker tissue samples were collected in 2011 in the Black Hills of South Dakota. Middle Rockies Ecoregion is shown in *light grey*, and Loss Zone is shown in *dark grey*. An *asterisk* corresponds to Annie Creek, where a small sample size was obtained

1000 spectrophotometer (Nanodrop Technologies, Wilmington, Delaware), then samples were frozen and shipped to the University of New Mexico for subsequent analysis.

Genetic variation was assessed across 10 microsatellite loci (Tranah et al. 2001; Turner et al. 2009; Dowling et al. 2012; Online Resource 1) to estimate genetic diversity and population structure. Forward primers were labeled with one of two fluorescent labeled dyes (FAM or HEX). Polymerase chain reaction (PCR) amplifications were performed under the following conditions: 5–10 ng DNA per µl (10 µl final reaction volume), 1X reaction buffer containing $Mg^{2+}$, 125 µM of each dNTPs, 0.4 µM of each primer, and 0.375 units of *Taq* DNA polymerase (GeneScript). Thermal cycling conditions were: initial denaturation (90 °C, 2 min); 30 cycles of denaturation (90 °C, 30 s), annealing (see Online Resource 1 for annealing temperatures, 30 s), elongation (72 °C, 45 s); and final elongation (72 °C, 15 min). Fragment lengths of PCR products were quantified on an automated DNA sequencer (ABI 3130; Applied Biosystems, Life Technologies, Carlsbad, CA) with HD400-ROX internal size standard (ABI) and analyzed with GeneScan software. Individuals were scored using GeneMapper software (ABI).

## Data analyses

### *Hardy–Weinberg equilibrium*

Samples from each site were tested for departure from Hardy–Weinberg equilibrium (HWE) using GENEPOP 4.1 (Raymond and Rousset 1995a; Rousset 2008). Significance values were computed using Fisher's exact test with a Markov Chain method using 100 batches and 1000 iterations each (Guo and Thompson 1992; Raymond and Rousset 1995b). Alpha values were adjusted for multiple comparisons using a sequential Bonferroni method (Rice 1989). Any locus or site not conforming to HWE expectations was specifically tested for heterozygote excess/deficiency, where departures were identified using a G-test implemented in GENEPOP 4.1 (Raymond and Rousset 1995a). Rare genotypes that are sometimes associated with microsatellite data can lead to significant deviations from

HWE using exact tests (Pamilo and Varvio-Aho 1984). Therefore, exact tests that identified statistically significant deviations from HWE expectations were re-tested following a procedure where genotypes with an expected frequency of less than 1 % were pooled and the locus re-tested using a Chi square goodness of fit test in Microsoft Excel (e.g., Hedrick 2000). Additionally, all pairs of loci were assessed for linkage disequilibrium using the Markov chain method with 1000 dememorizations, 100 batches and 1000 iterations per batch in GENEPOP 4.1. Microsatellite loci were screened for large allele dropout, null alleles, and scoring errors that could result from stuttering using MICRO-CHECKER (van Oosterhout et al. 2004).

*Genetic diversity*

Microsatellite diversity within and between sample sites was compared using allele frequency distributions, observed ($H_O$), and expected ($H_E$) heterozygosities. Additionally, allelic richness estimates ($A_r$) and private allele estimates were made using the rarefaction method described by Goudet (1995) and Petit et al. (1998) as recommended by Leberg (2002) to account for unequal sample sizes using HP-RARE (Kalinowski 2005). Private allele frequencies were calculated using GENEPOP. Simple linear regression was used to assess the relationships between stream fragment length on both allelic richness and expected heterozygosity.

*Population structure*

Spatial structure was assessed using the Weir and Cockerham (1984) estimator of global $F_{ST}$ calculated in FSTAT (Goudet 2001). Confidence intervals were calculated using a jackknife approach in FSTAT. Pairwise estimates of $F_{ST}$ were used to quantify genetic differentiation among sites and were calculated in Arlequin (ver. 3.5.1.2, Excoffier and Lischer 2010). Pairwise $F_{ST}$ values were tested for statistical significance via permutation (1000 replicates) in Arlequin. We also used a Bayesian approach as implemented in the program STRUCTURE (Pritchard et al. 2000) to assess genetic structuring across sites. To test whether structure existed among streams we evaluated our data for $K = 1$–5 (5 iterations per $K$ value); where $K$ is the number of potential genetic units (i.e., streams) occurring within the dataset. We used a 250,000 replicate burn-in period followed by 250,000 replicates per iteration. We used the admixture model with and without locations as prior probabilities and correlated allele frequencies. Discrete genetic units were identified by visual examination of STRUCTURE plots, and by incorporation of the Evanno method (Evanno et al. 2005) as implemented in STRUCTURE HARVESTER (Earl and vonHoldt 2012; Accessed 6/28/2013). To test for hierarchical genetic structuring between sites in tributaries of the Belle Fourche and Cheyenne rivers we used analysis of molecular variance (AMOVA) performed in Arlequin. Significance levels for AMOVA were computed using 1000 non-parametric permutations of the data. Additionally, AMOVA was performed using genetic groups indicated by the program STRUCTURE. Finally, we tested for isolation by distance by using a Mantel test to evaluate the correlation between pairwise $F_{ST}$ values and pairwise stream-course distance (km) between sites (estimated in Google Earth). The Mantel test was performed using R (ver. 2.15.0, R Development Core Team, 2012) with the package *vegan* (Oksanen et al. 2013).

*Genetic effective size*

Genetic effective size, $N_{eD}$, was estimated by the linkage disequilibrium method (Hill 1981) using the program LDNE (Waples and Do 2008). $N_{eD}$ was estimated for each stream separately and using all individuals combined ($n = 185$) for a global $N_{eD}$ because the linkage disequilibrium method is considered to be robust in structured populations (Waples and England 2011). Alleles with low frequencies can bias estimates of $N_{eD}$ (Waples and Gaggiotti 2006); therefore we excluded alleles with frequencies less than or equal to 0.03, as suggested by Waples and Do (2010). Upper and lower 95 % confidence intervals for $N_{eD}$ were calculated using a jackknife approach implemented in LDNE.

## Results

Genetic tissue samples were collected from a total of 187 mountain suckers and six streams (Table 1). A minimum of 30 fish were clipped from each stream, with the exception of Annie Creek, where only two individuals were collected. Thus, the Annie Creek population was not used in analyses of genetic structure.

### Hardy–Weinberg equilibrium

Initially, 10 of 60 locus-by-site comparisons had significant deviations from HWE. Following sequential Bonferroni correction of nominal alpha, five of 60 comparisons still yielded significant deviations from HWE. However, after pooling rare (<0.01) expected genotypes, only one stream population showed a significant deviation from HWE, Boxelder Creek at locus *Dlu4184*. No significant linkage disequilibrium was observed between any pairs of loci. Forty-eight of 50 locus-by-site comparisons were non-significant for heterozygote deficiency; exceptions were Whitewood Creek at locus *Dlu233* and locus *Xte9*. No locus-by-site comparisons yielded

**Table 1** Sample size (N) and genetic diversity statistics for populations of mountain sucker collected in 2011 in the Black Hills of South Dakota, USA

| Stream | N | Loci typed | $H_E$ | $H_O$ | $A_R$ | Private $A_R$ | $F_{IS}$ |
|---|---|---|---|---|---|---|---|
| Boxelder | 33 | 10 | 0.656 | 0.607 | 6.91 | 1.05 | 0.076 |
| Bear Butte | 33 | 10 | 0.605 | 0.583 | 5.67 | 0.31 | 0.036 |
| Elk | 49 | 10 | 0.621 | 0.612 | 6.09 | 0.37 | 0.014 |
| NF rapid | 30 | 10 | 0.623 | 0.603 | 6.37 | 0.70 | 0.034 |
| Whitewood | 40 | 10 | 0.644 | 0.603 | 6.41 | 0.51 | 0.064 |

Summary statistics include Nei's unbiased gene diversity ($H_E$), observed heterozygosity ($H_O$), allelic richness ($A_R$), private $A_R$, and inbreeding coefficient ($F_{IS}$) tabulated by stream. A dash indicates that no estimate could be made due to small sample size

significant heterozygote excess values. Three loci (*Dlu233*, *Dlu4184*, and *Xte9*) were found to have an excess of homozygotes, which could indicate presence of null alleles. Xte9 exhibited an excess of homozygotes at all sites except for North Fork Rapid Creek; however, removal of *Xte9* had no appreciable effect on results.

## Genetic diversity

Mountain sucker samples had similar levels of $H_E$ and $H_O$ across all five streams with adequate sample sizes (range 0.605–0.656 and 0.583–0.612, respectively; Table 1). Boxelder Creek had the highest allelic richness and private allelic richness ($A_R = 6.91$, private $A_R = 1.05$), whereas Bear Butte Creek had the lowest ($A_R = 5.67$, private $A_R = 0.31$). Inbreeding coefficients ($F_{IS}$) were greatest in Boxelder Creek ($F_{IS} = 0.076$; Table 1) and Whitewood Creek ($F_{IS} = 0.064$), and lowest in Elk Creek ($F_{IS} = 0.014$). There was no statistically significant relationship between either allelic richness or expected heterozygosity and stream fragment length (Fig. 2; $A_R$: $r^2 = 0.25$, $P = 0.39$; $H_E$: $r^2 = 0.50$, $P = 0.12$). Despite being non-significant, there appear to be trends. Mountain suckers in the longest stream fragment exhibited high allelic richness (Whitewood Creek; 48.3 km), whereas allelic richness was low in the shortest stream fragment (Bear Butte Creek; 13 km). Excluding Boxelder Creek, 4 of the 5 sampled stream fragments varied from 22.2 to 48.3 km in total length and observed heterozygosities ranged between 0.603 and 0.612. The shortest wetted stream fragment occupied by mountain suckers (Annie Creek; 2 km) had the lowest observed heterozygosity (0.400), and the second shortest stream fragment (Bear Butte Creek; 13 km) had the second lowest observed heterozygosity (0.583). Boxelder Creek was intermediate in fragment length (26 km), but mountain suckers there had the greatest expected heterozygosity (0.656) and the greatest allelic richness (6.91).

## Population structure

Comparisons of five streams (where $n \geq 30$) indicated that there was significant genetic structuring in the Black Hills

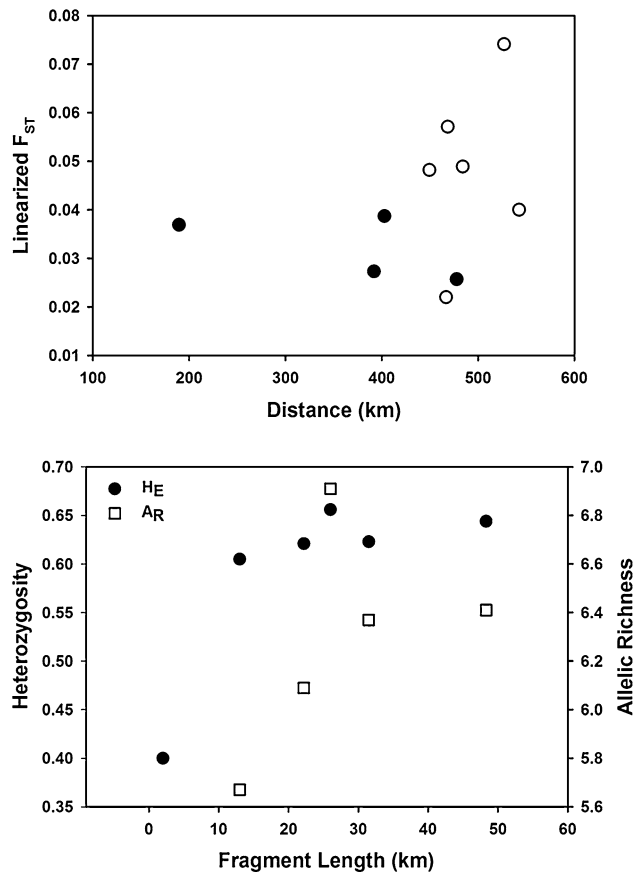


**Fig. 2** Pairwise $F_{ST}$ plotted by stream distance between sites from which mountain suckers were collected in 2011 in the Black Hills of South Dakota (*top panel*). Comparisons between Belle Fourche and Cheyenne River drainages are indicated with open circles and comparisons within drainages indicated by *closed circles*. Expected heterozygosity ($H_E$) and allelic richness ($A_R$) plotted against stream fragment length (*bottom panel*). Annie Creek ($n = 2$ individuals genotyped) is included in this plot (i.e., $H_E = 0.400$) for reference, but interpretation of those data are limited to the small number of individuals collected

(Global $F_{ST} = 0.041$, 95 % CI 0.022–0.061). Populations in Bear Butte Creek and the North Fork of Rapid Creek were most divergent and the Elk Creek and Whitewood Creek populations were least divergent (Table 2). The STRUCTURE model without sample locations as prior probabilities suggested the most likely number of unique genetic groups

was K = 1 based on visual inspection of STRUCTURE plots and low ΔK for K > 1 in the Evanno table (Online Resource 1). Given the significant allelic differentiation among streams we used 'streams' as prior probabilities to help infer genetic structure (Hubisz, et al. 2009). Results of the 'locations as priors' model indicated support for two genetic units (K = 2) and for all streams harboring a distinct genetic unit (K = 5; Fig. 3, Online Resource 1). Under K = 2, STRUCTURE suggested Bear Butte, Elk and Whitewood creeks were one genetic unit, and Boxelder and North Fork Rapid creeks were another genetic unit (Fig. 3). AMOVA failed to resolve hierarchical genetic structure when streams were grouped into Belle Fourche or Cheyenne drainages because the majority of variance was explained by differences among populations (95.7 % variation, $F_{ST} = 0.043$, $P < 0.001$), with a small but significant portion of variance explained among drainages (3.6 %, $F_{CT} = 0.036$, $P < 0.001$) (Table 3). Grouping samples by genetic units suggested by STRUCTURE (K = 2) also resulted in the majority of variance explained by differences among individuals within populations (95.2 %, $F_{ST} = 0.048$, $P < 0.001$) with little variation explained among groups (1.5 %, $F_{CT} = 0.015$, $P = 0.016$) or among populations within groups (3.3 %, $F_{SC} = 0.033$, $P < 0.001$). Correlation among stream distance and pairwise $F_{ST}$ was not significant (Mantel r = 0.30, P = 0.20; Fig. 2).

### Genetic effective size

Estimates of $N_{eD}$ within streams were generally unreliable because LDNE either produced negative or very large estimates and confidence intervals were infinite (Boxelder $N_{eD} = 1238$ [101–∞ 95 % CI]; Bear Butte $N_{eD} = -201$ [215–∞]; Elk $N_{eD} = 211$ [85–∞]; North Fork Rapid $N_{eD} = -458$ [157–∞]; Whitewood $N_{eD} = -1404$ [135–∞]). Global $N_{eD}$ of all streams pooled (n = 185) was 338 individuals (95 % CI 194–932).

## Discussion

This study was the first to describe fine-scale genetic diversity and spatial genetic structure in a disjunct peripheral population of mountain sucker in the Black Hills of South Dakota. This population (i.e., the entire collection of individuals sampled in the Black Hills of SD for this study), isolated on the northeast edge of the native range, may be more valuable to conservation of the species than populations nearer the core of the range (Bunnell et al. 2004; Osborne et al. 2012), although direct genetic comparisons with core populations were not included in this study. Divergence of disjunct peripheral populations from core populations can result from founder effects (Carson 1971), genetic bottlenecks (Nei et al. 1975), and genetic drift (Nei et al. 1975). Potential for bottlenecks, reduced gene flow, and local adaptation is sometimes greater in peripheral populations than in contiguous or core populations (Lesica and Allendorf 1995).

**Table 2** Pairwise $F_{ST}$ values among mountain sucker populations in 5 Black Hills streams in South Dakota, USA

| | Boxelder | Bear Butte | Elk | NF rapid |
|---|---|---|---|---|
| Bear Butte | 0.057 | | | |
| Elk | 0.039 | 0.048 | | |
| NF rapid | 0.027 | 0.074 | 0.026 | |
| Whitewood | 0.049 | 0.037 | 0.022 | 0.040 |

All pairwise $F_{ST}$ P values < 0.001

### Standing stocks of genetic diversity in existing stream fragments

Genetic drift is likely the predominant force shaping contemporary, local genetic diversity of mountain suckers in the Black Hills of South Dakota. This is because the efficacy of genetic drift as an evolutionary force increases as a population becomes smaller and more isolated. Since the early 1900s, the water 'loss zone' encircling the Black Hills (Williamson and Carter 2001), fire suppression (i.e., increased transpiration and decreased surface flow; Koth 2007), severe droughts (i.e., 1930s, 1950s, and 1988; Bunkers et al. 1999), dewatering for gold mining activities (Koth 2007), and dam construction (i.e., 1930s Works Progress Administration and Civilian Conservation Corps projects; Chipps et al. 2007), have resulted in fragmentation and reduced gene flow among streams within the Black Hills. Higher-order prairie rivers, such as the Belle Fourche and Cheyenne rivers, may also pose a seasonal barrier to gene flow because summer water temperatures in these rivers regularly exceed 31.5 °C, which is the critical thermal maximum for mountain sucker (Schultz and Bertrand 2011). Fisheries surveys and historical distribution comparisons indicate that this disjunct peripheral population is relatively small in demographic terms (Dauwalter and Rahel 2008; Schultz and Bertrand 2012), and after four decades of consistent decline, possibly continuing to shrink. The relatively low estimate of global $N_{eD} = 338$ is consistent with findings based on demographic data, although genetic information on core populations is needed to fully interpret this finding.

In general, small, isolated population fragments are often associated with lowered levels of genetic diversity when compared to large, continuously distributed populations. In this study, as stream fragment length increased, allelic richness and observed and expected heterozygosity appeared to increase, although the correlations were not statistically significant at $\alpha = 0.05$. Additional variation in genetic diversity may be partially explained by differences

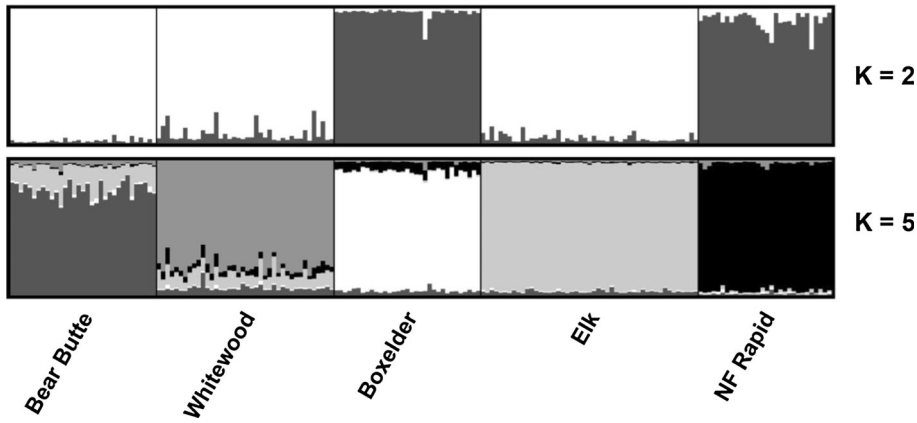


**Fig. 3** STRUCTURE results for mountain suckers collected in 2011 in the Black Hills of South Dakota. The colors of each individual *bar* represent the proportional assignment of the individual into K genetic clusters

**Table 3** Analysis of molecular variance (AMOVA) for populations of mountain sucker collected in 2011 from the Black Hills of South Dakota, USA

| Source of variation | SS | Variance component | % variation |
|---|---|---|---|
| Drainage basin | 17.8 | 0.04 | 1.1 |
| Pops w/in basins | 33.5 | 0.12 | 3.5 |
| Within pops | 1119.2 | 3.15 | 95.5 |
| Total | 1170.4 | 3.30 | |
| STRUCTURE groups | 18.4 | 0.05 | 1.5 |
| Pops w/in groups | 32.9 | 0.11 | 3.3 |
| Within pops | 1119.2 | 3.15 | 95.2 |
| Total | 1170.4 | 3.30 | |

Populations were grouped by drainage basin (Cheyenne River or Belle Fourche River) or by genetic units indicated by STRUCTURE

in habitat suitability and quality across stream fragments. High quality habitat can support higher densities of fish in smaller stream fragments. Mountain sucker density varies markedly among Black Hills streams. For example, Elk Creek had mountain sucker densities of over 3000 fish per hectare (ha), which is 1.7 times higher than the next most dense population in Whitewood Creek (1762 fish/ha). The Elk Creek drainage is narrow, encompassing only the immediate floodplain and riparian area of Elk Creek itself, thus the habitat and water quality are good and strongly dependent on land use in a small area (Koth 2007). In contrast, Whitewood Creek, with over 48 km of wetted stream length occupied by suckers has been heavily polluted by mining activities, and the habitat and water quality is compromised, which may explain the intermediate allelic richness and observed heterozygosity.

## Population structure

Overall low levels of genetic structure suggest streams in this study were connected by moderate to high levels of gene flow in the past century. Gene flow now appears to be restricted among streams as a result of changes in wetted stream area. Gene flow among mountain suckers inhabiting different streams was generally restricted to historical high flows that sustained surface flow across the 'loss zone' in streams within either the Belle Fourche or the Cheyenne River drainage; the 'locations as priors' model in STRUCTURE generally grouped streams within drainages. However, Elk Creek clustered with other tributaries of the Belle Fourche drainage. It is possible that this is a result of stream capture or anthropogenic transfer (e.g., bait bucket or creel) between Elk and Bear Butte Creek, since the headwaters of these two streams are less than 3 km apart (Fig. 1). Pairwise $F_{ST}$ values confirmed that there were small, but significant differences between all streams in the Black Hills. This suggests that, at present, genetic drift is more important than gene flow in determining the genetic population structure of mountain sucker in the Black Hills of South Dakota.

## Conservation implications of genetic diversity and connectedness

The trend of lower genetic diversity in shorter stream fragment lengths documented here, coupled with declines in occupancy and density reported in Schultz and Bertrand (2012), suggest that Black Hills mountain sucker genetic diversity could continue to decline with increasing stream fragmentation. Rates of decline and loss could increase as a result of a negative feedback loop between inbreeding and viability (i.e., extinction vortex; Gilpin and Soulé 1986). However, there is little evidence of inbreeding within streams in the Black Hills. There were notable increases in allelic richness and observed heterozygosity when stream fragment length increased from 13 to 22.2 km (Bear Butte and Elk creeks), and at 26 km (Boxelder Creek) and longer (North Fork Rapid and Whitewood creeks), but stream fragment length was not a significant predictor of allelic

richness or observed heterozygosity. Habitat quality might be more influential.

Two conservative management approaches that could help stem the loss of genetic diversity are (1) to prevent further stream fragmentation and (2) to increase stream fragment lengths, habitat quality, and connectivity by remediating or removing barriers to fish migration. Brown trout and brook trout overlap the mountain sucker distribution throughout much of the Black Hills. Schultz and Bertrand (2011) found high densities of mountain sucker at reaches exhibiting relatively low densities of trout (<0.1 fish/m$^2$). Non-native salmonids have been introduced to nearly every reach of stream in the Black Hills due to widespread stocking and illegal introductions (i.e. "bait bucket introductions"). Therefore, the current distribution and density of nonnative salmonids in the Black Hills appears to be limited by stream characteristics, rather than barriers. We suspect that there also could be a 'point of no return' fish density subject to the Allee effect (Allee and Bowen 1932), but to our knowledge, this has not been demonstrated yet in this species. Furthermore, the Black Hills mountain sucker population is so limited in distribution and density that we were unable to generate an estimate of effective population size for each stream sampled. An alternative explanation is that local population sizes are larger than can be detected with the sample sizes obtained for the purposes of this study. This result also is consistent with low inbreeding within streams. Some Black Hills streams may contain reasonably good standing levels of genetic diversity, but since they have now become isolated, we anticipate these levels to diminish over time. If responsible agencies (e.g., United States Forest Service) strive to maintain or restore the Black Hills mountain sucker population, then regular demographic monitoring and periodic genetic monitoring will provide fisheries biologists with reliable census data on which to base their management decisions.

Management actions that are more likely to increase densities and genetic diversity of mountain sucker and other Black Hills fish populations are habitat conservation, restoration of previously occupied habitat, enhancement of habitats currently supporting low densities of mountain sucker, and repatriation of previously occupied habitat (since diversity remains intact in streams at the moment). Small areas of high habitat quality and high fish density would provide the greatest return for habitat conservation investments and might also serve as donor populations for repatriation of previously occupied stream fragments. Elk Creek was the most densely occupied mountain sucker habitat in the Black Hills (Schultz and Bertrand 2012), and thus could be preserved in its present state. Whitewood Creek, which flows continuously, even through the 'loss zone' (Koth 2007), has the potential to provide more abundant perennial habitat to mountain sucker but habitat and water quality would need to be improved to increase mountain sucker densities. Another conservation opportunity is population restoration into previously occupied habitats in which water quality (remediation of iron and/or cyanide pollution), physical habitat (channel modification to reverse siltation), and biotic community (avoid negative interactions with non-coevolved predators) have been enhanced to promote success of native fishes. Habitat planning should take into consideration the potential effects of climate change on mountain sucker habitat suitability: thermal tolerances may shift mountain sucker and other fishes upstream into smaller tributary habitats (e.g., Isaak et al. 2010).

Maintaining or enhancing mountain sucker genetic diversity in the Black Hills is possible if managers prioritize (1) increasing the number of occupied streams, and (2) increasing fish density in occupied streams. A conservation hatchery program for this species might also be warranted to help maintain diversity in case of catastrophic loss of high diversity localities in the wild. However, stocking from the hatchery should be used prudently and sparingly to avoid negative effects associated with hatchery-induced selection (Araki and Schmid 2010). Mountain suckers historically occupied 33 stream reaches in 26 named streams throughout the Black Hills and now occur in only 11 stream reaches that are each in a different named stream. In five of the currently occupied reaches, fish densities have decreased by over 80 %. We found that genetic drift within streams is likely occurring, but there is evidence that gene flow among tributaries of the Belle Fourche River and among tributaries of the Cheyenne River likely occurred within the past century. Therefore, we recommend that if managers decide to repatriate mountain suckers, they use dense source populations such as Elk Creek. It also may be prudent to artificially restore gene flow to small, isolated population fragments, such as Annie Creek.

Small, disjunct peripheral populations, like the mountain sucker population in the Black Hills of South Dakota, are at greater risk of extirpation than continuous populations at the core of a species range (Hewitt 1996; Bunnell et al. 2004). However, peripheral populations also may contain the necessary genetic diversity and individuals with traits best suited to colonize new habitats in response to climate change and other anthropogenic affects (Hewitt 1999; Comte and Grenouillet 2013). The combination of stream survey and genetic data for the Black Hills streams suggest that there are numerous opportunities for management aimed at enhancing persistence of local diversity in the face of large-scale environmental change. We anticipate that mountain sucker will be an interesting test case for future demo-genetic (sensu Paz-Vinas et al. 2013) studies

and management of similarly small and shrinking disjunct peripheral populations of native fishes, particularly suckers (Family Catostomidae; Marlis Douglas pers. comm.).

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